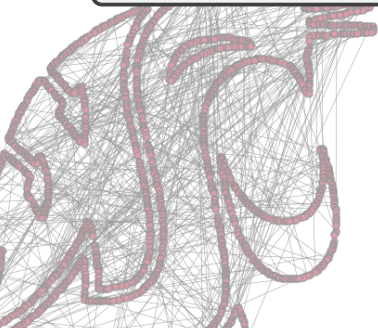


A Machine Learning Approach to Catalysis Tuning

Garrett Kepler

WNAR/IBS
June 16th, 2026



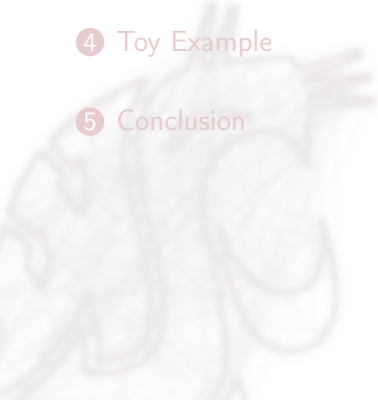
Collaboration

The following presentation discusses a modeling technique inspired by the theoretical work I am doing on a project with Raul Pelaez Garcia in the Chemical Engineering department here at WSU.



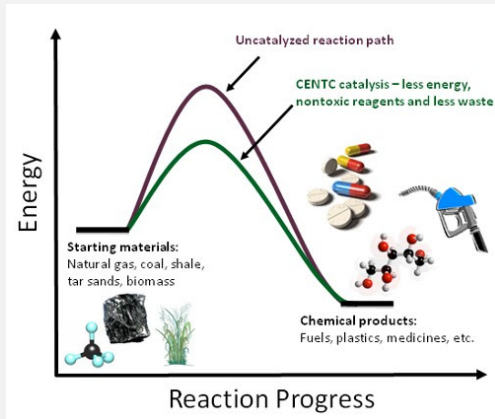
Outline

- ① Catalysis Basics
- ② Modeling Problem
- ③ GVAEs
- ④ Toy Example
- ⑤ Conclusion

- 
- 1 Catalysis Basics
 - 2 Modeling Problem
 - 3 GVAEs
 - 4 Toy Example
 - 5 Conclusion

Chemistry Background

The idea of *catalysis* is to complete chemical reactions using less time and/or energy by adding another chemical called a *catalyst*.



Chemistry Background

One great example of catalysis in action is the *catalytic* converter most cars have today.

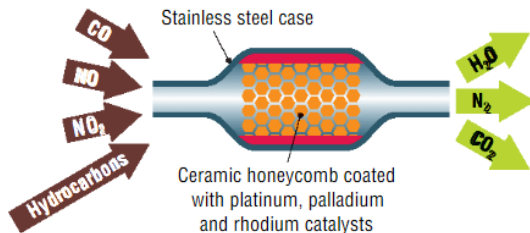
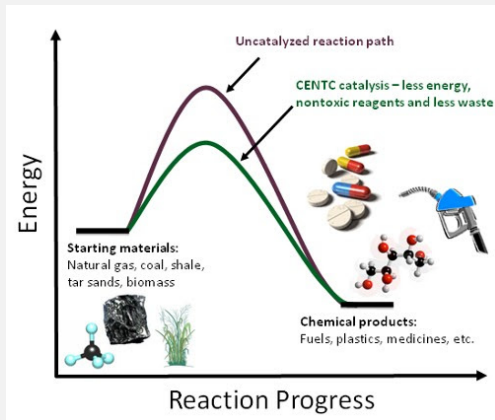
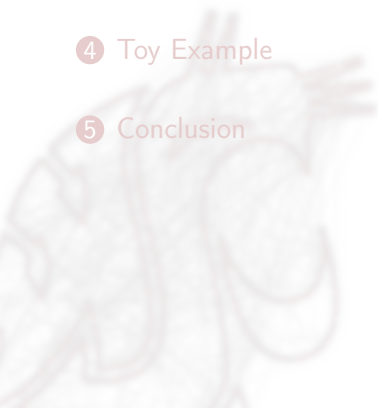


Fig. 16.32 The main gases entering a catalytic converter and leaving the catalytic converter.

Chemistry Background

However, these theoretical curves are idealized scenarios that are not used often in practice. Tuning of systems is instead done by brute force. That is, changing system parameters until satisfied.



- 
- 1 Catalysis Basics
 - 2 Modeling Problem**
 - 3 GVAEs
 - 4 Toy Example
 - 5 Conclusion

Geometry of Our Filters

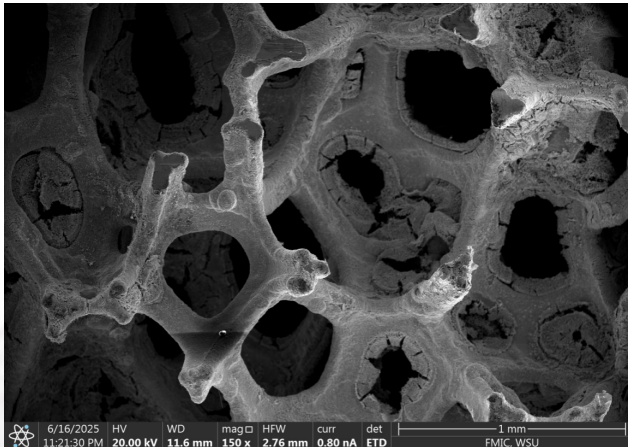


Figure: A cross-section of the filters we are dealing with in the project. Note there is some regularity in the geometry, but flaws exist.

Geometry of Our Filters

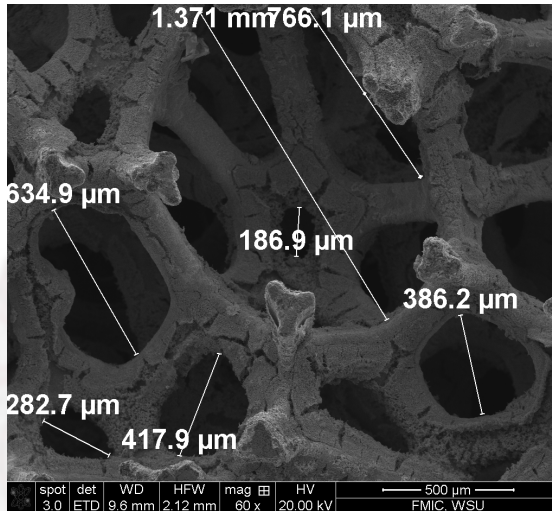


Figure: These filters are incredibly fine.

The Goal

In short:

Money and energy can be saved by simulation. But, existing programs are typically computationally involved fluid dynamics programs that require full scans of filters and many samples to be useful.

A principled approach to get an idea of where to set system parameters could speed up the tuning process.

Why Graphs

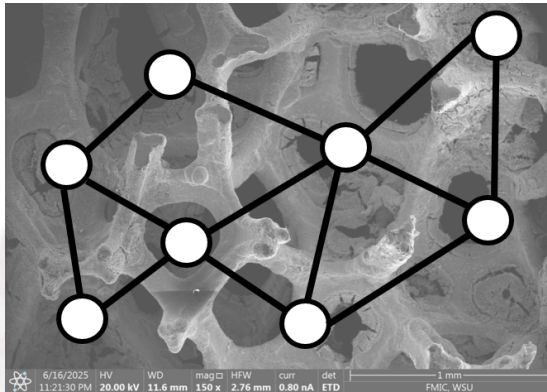
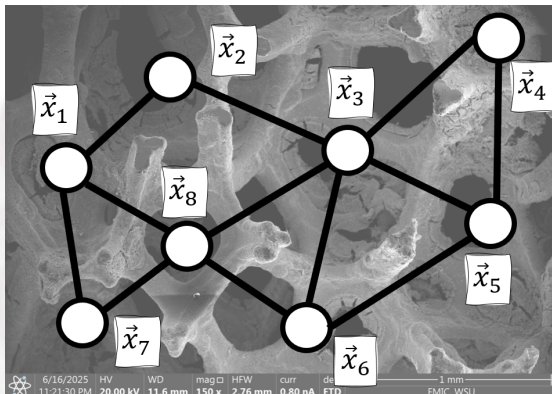
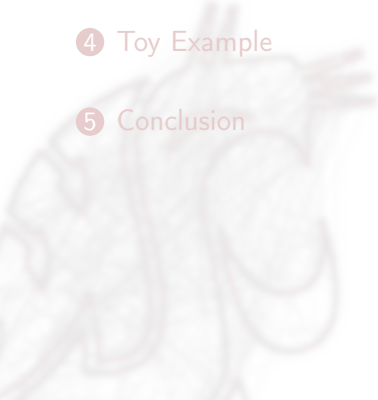


Figure: An example of the graph approximation. Nodes correspond to pockets in the filter with edges connecting adjacent pockets.

Why Graphs

Analyzing what happens *within* the filter is a big part of understanding how to tune systems. For example, the reaction efficiency is highly dependent on temperature and magnetization of the pores within.

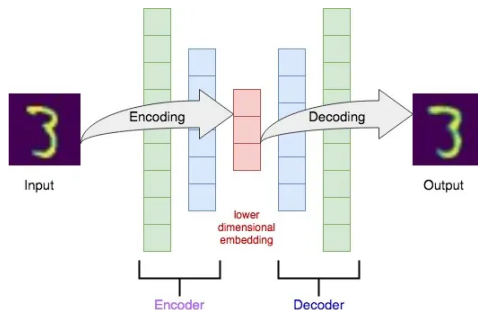


- 
- 1 Catalysis Basics
 - 2 Modeling Problem
 - 3 GVAEs**
 - 4 Toy Example
 - 5 Conclusion

Auto-Encoders

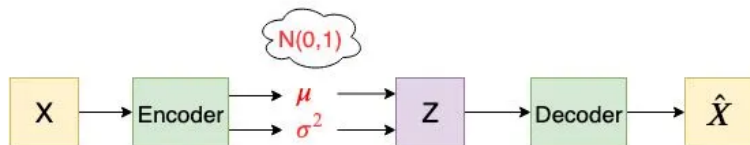
A traditional auto encoder learns to do two things:

1. Effectively encode by taking inputs and embedding them in a lower dimension
2. Effectively decode by taking the low dimensional embeddings back to the higher dimension



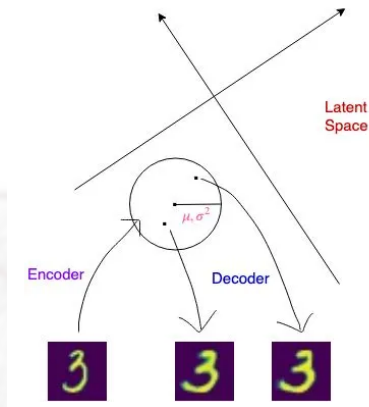
Variational Auto-Encoders

Rather than assigning each input to a *single point* in a lower dimension, variational auto-encoders assign the input a *probability distribution* in the latent space to sample an embedding from.



Variational Auto-Encoders

Our decoder will now be learning to reconstruct the outputs from a *random sample* rather than a deterministic point.



Our VAE: Architecture

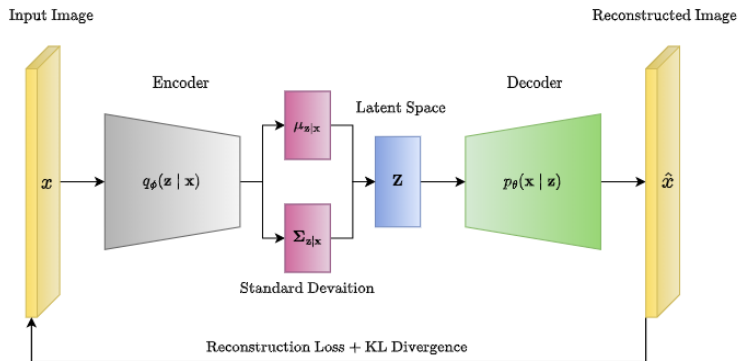


Figure: A typical variational auto-encoder architecture. Graph-VAEs follow this same structure. In the seminal (3-page) paper on the topic, Kipf and Welling introduced a framework to follow: <https://arxiv.org/abs/1611.07308>.

Our VAE: Specifics

Forward Pass:

- Input: Take a graph G on n nodes with corresponding $\{0, 1\}$ adjacency matrix $A \in \mathbb{R}^{n \times n}$. Vectorize A into $\vec{x} \in \mathbb{R}^{n^2}$.
- Encode: Send $\vec{x}$ through two-layer MLP to determine mean vector ($\vec{\mu}$) and variance ($\log \vec{\sigma}^2$) in latent space $\mathbb{R}^m$.
- Sample Embedding: Sample $\vec{\epsilon} \sim \mathcal{N}(\vec{0}, \mathbb{I})$ and set $\vec{z} = \vec{\mu} + \vec{\sigma} \circ \vec{\epsilon}$
- Decode: Send $\vec{z}$ through two-layer MLP to obtain $\vec{y} \in \mathbb{R}^{n^2}$ and apply sigmoid function to $\vec{y}$ to obtain a vector of probabilities $\hat{x} \in \mathbb{R}^{n^2}$ and form $\hat{A} \in \mathbb{R}^{n \times n}$.
- Generate Graph: Define a threshold (say, .5) and form a $\{0, 1\}$ matrix $A' \in \mathbb{R}^{n \times n}$ where $A'_{ij} = 1$ if $\hat{A}_{ij} \geq .5$ and 0 otherwise.

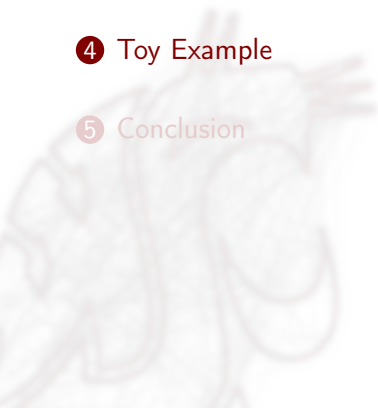
Our GVAE: Specifics

Backward Pass:

- In the end, we use standard backpropagation penalizing for bad reconstruction, $\mathcal{L}_{\text{recon}}$, and penalizing for distributional shifts in $\vec{z}$, $\mathcal{L}_{KL}$.

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{\text{recon}} + \mathcal{L}_{KL} \\ &= - \sum_{i=1}^{n^2} [x_i \log(\hat{x}_i) + (1 - x_i) \log(1 - \hat{x}_i)] \\ &\quad - \frac{1}{2} \sum_{k=1}^d (1 + \log \sigma_k^2 - \mu_k^2 - \sigma_k^2)\end{aligned}$$

where d is the latent dimension ($\vec{z} \in \mathbb{R}^d$)

- 
- 1 Catalysis Basics
 - 2 Modeling Problem
 - 3 GVAEs
 - 4 Toy Example**
 - 5 Conclusion

Erdős-Renyi Random Graphs

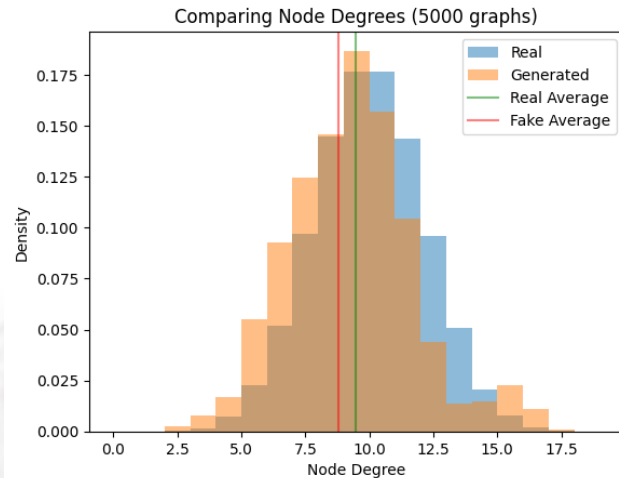


Figure: Degree distribution of 5000 GVAE generated graphs versus 500 training Erdős-Renyi Graphs

Erdős-Renyi Random Graphs

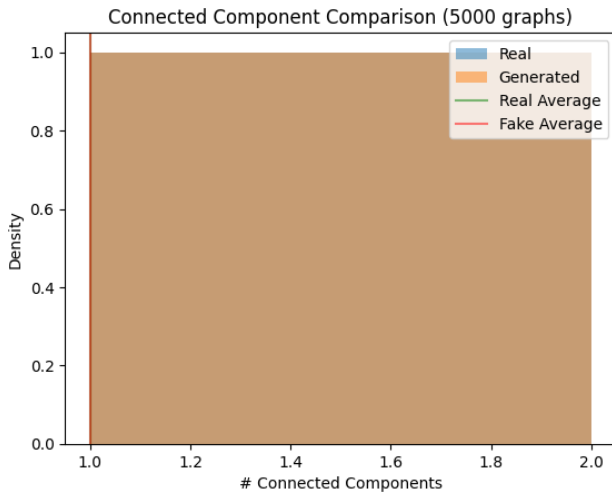


Figure: The GVAE learned to preserve connectedness

Erdős-Renyi Random Graphs

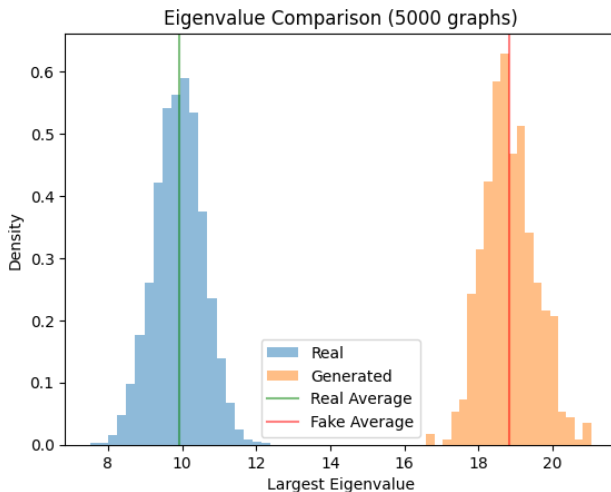


Figure: But, the graphs are not isomorphic to the training graphs.

Erdős-Renyi Random Graphs

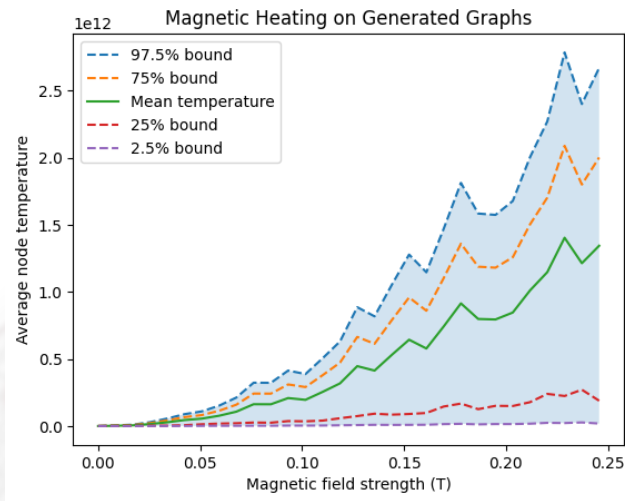
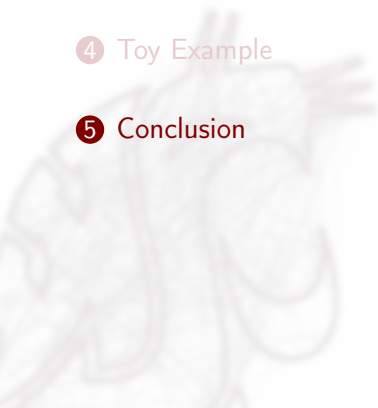


Figure: Behavior of GVAE generated graph temperature depending on magnetic field

- 
- ① Catalysis Basics
 - ② Modeling Problem
 - ③ GVAEs
 - ④ Toy Example
 - ⑤ Conclusion**

To Summarize

Overview:

- We can use GVAEs to generate “similar” graphs
- With a collection of graphs informed by actual filter data, we can train a GVAE to generate “similar” filter structure representations
- We can approximate how the filter will behave in some situation before wasting time and money tuning

Question?

Thank you!
garrett.kepler@wsu.edu

Photo Sources

- Catalysis curve source:
<http://depts.washington.edu/centc/catalysis.php>
- Catalytic converter layout: <https://cataler.com/catalyst/>